

Conjecture 00000007783: the cube has thin-shell standard deviation at least $1/\sqrt{15}$ in every dimension, so σ_n is not $O(n^{-1/4})$

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1 The conjecture

English. Definition: The thin-shell constant σ_n is the supremum over isotropic convex bodies of the standard deviation of $|X|$ for an isotropic uniform sample. Conjecture: $\sigma_n = \Theta(n^{-1/4})$, so the known upper-bound order cannot be improved; and the σ values of balls and simplices are $n^{-1/2}$ and $cn^{-1/4}$, all isotropic bodies interpolating between the two strictly monotonically, with interpolation law given by the volume-concentration exponent of a two-parameter random simplex family.

Chinese text. The Chinese version says the same. Its definition reads (translated) “ σ_n is the supremum, over all isotropic convex bodies, of the standard deviation of $|X|$ (the norm of an isotropic sample point of the uniform convex body)”, and its interpolation clause reads “the thin-shell constants of all isotropic bodies lie between the two”.

2 Reading and conventions

Work in $\mathbb{R}^n$ with the Euclidean norm $|x|$ and Lebesgue measure vol (Lean: `EuclideanSpace` $\mathbb{R}$ (Fin n) and `volume`).

Definition 1. A *convex body* is a compact convex set $K \subseteq \mathbb{R}^n$ with nonempty interior. The *uniform law* on K is $\text{vol}(K)^{-1} \text{vol}|_K$ (Lean: `volume[|K|]`, Mathlib’s `ProbabilityTheory.cond`). K is *isotropic* if $X \sim \text{unif}(K)$ satisfies $\mathbb{E}X_i = 0$ and $\mathbb{E}X_i X_j = \delta_{ij}$ for all i, j , i.e. mean 0 and covariance I (the text’s “isotropic uniform sample”). The *thin-shell quantity* is $\sigma(K) = \sqrt{\text{Var}|X|}$ (Mathlib `variance`). The thin-shell constant is $\sigma_n = \sup_K \sigma(K)$ over isotropic convex bodies $K \subseteq \mathbb{R}^n$, taken in $[0, \infty]$.

The conjecture is a conjunction. $\sigma_n = \Theta(n^{-1/4})$ contains the upper bound: there are C, N with $\sigma_n \leq Cn^{-1/4}$ for all $n \geq N$. The interpolation clause says that every isotropic body $K \subseteq \mathbb{R}^n$ has $\sigma(K)$ between the ball value $n^{-1/2}$ and the simplex value $cn^{-1/4}$, so in particular $\sigma(K) \leq cn^{-1/4}$ for one fixed $c > 0$. **We refute exactly these two clauses:** the upper bound in $\sigma_n = \Theta(n^{-1/4})$, and the upper half of the interpolation clause. Here $n^{-1/4}$ is the real power $n^{-1/4} = \exp(-\frac{1}{4} \log n)$ for $n \geq 1$.

Not assessed. We do not use or formalize the ball value, the simplex value, the strict monotonicity, or the “volume-concentration exponent” law. (For the record, the isotropic ball of radius $\sqrt{n+2}$ has $\text{Var}|X| = n/(n+1)^2$, which is of order n^{-1} ; this is not used.)

Volume-one convention. Some authors call K isotropic when $\text{vol}(K) = 1$, the barycenter is 0 and the covariance is $L_K^2 I$ for some $L_K > 0$. We prove the same conclusion in that convention as well (Theorem 7(c)).

3 The cube

Fix $a > 0$ and let $Q_a = [-a, a]^n$. Let X be uniform on Q_a .

Lemma 2. Q_a is a convex body, $\text{vol}(Q_a) = (2a)^n$, and $|x|^2 \leq na^2$ on Q_a .

Proof. Convexity is coordinatewise. Q_a is closed and lies in the closed ball of radius $\sqrt{n}a$, so it is compact. It contains the open ball of radius a about 0, because $|x_i| \leq |x|$. The volume is a product of interval lengths, and $|x|^2 = \sum_i x_i^2 \leq na^2$. $\square$

Lemma 3. Let $c_m = (2a)^{-1} \int_{-a}^a t^m dt$, so $c_0 = 1$, $c_1 = 0$, $c_2 = a^2/3$ and $c_4 = a^4/5$. For every exponent vector $p \in \mathbb{N}^n$, $\mathbb{E} \prod_i X_i^{p_i} = \prod_i c_{p_i}$. Hence $\mathbb{E} X_i = 0$, $\mathbb{E} X_i X_j = \delta_{ij} a^2/3$, $\mathbb{E} X_i^2 X_j^2 = a^4/9$ for $i \neq j$ and $\mathbb{E} X_i^4 = a^4/5$.

Proof. The map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ from Euclidean space to the product space preserves Lebesgue measure, and $\text{vol}|_{Q_a}$ becomes the product of the restrictions of Lebesgue measure to $[-a, a]$. Fubini's theorem for products gives $\int_{Q_a} \prod_i x_i^{p_i} dx = \prod_i \int_{-a}^a t^{p_i} dt = (2a)^n \prod_i c_{p_i}$, and we divide by $\text{vol}(Q_a) = (2a)^n$. The values of c_m follow from $\int_{-a}^a t^m dt = (a^{m+1} - (-a)^{m+1})/(m+1)$. $\square$

Lemma 4. Let $R = |X|$. Then $\mathbb{E} R^2 = na^2/3$ and $\mathbb{E} R^4 = n^2 a^4/9 + n(a^4/5 - a^4/9)$, so $\mathbb{E} R^4 - (\mathbb{E} R^2)^2 = \frac{4}{45} na^4$.

Proof. $R^2 = \sum_i X_i^2$ and $R^4 = \sum_{i,j} X_i^2 X_j^2$; apply Lemma 3 and linearity. $\square$

Theorem 5. For $n \geq 1$, $\text{Var}|X| \geq a^2/45$.

Proof. Let $m = \mathbb{E} R$ and $b = \sqrt{n}a$. By Lemma 2, $0 \leq R \leq b$ almost surely, so $0 \leq m \leq b$ and $R+m \leq 2b$. Pointwise, $(R^2 - m^2)^2 = (R - m)^2(R + m)^2 \leq 4b^2(R - m)^2$, hence $\mathbb{E}(R^2 - m^2)^2 \leq 4b^2 \text{Var} R$. On the other hand, $\mathbb{E}(R^2 - m^2)^2 = \mathbb{E} R^4 - 2m^2 \mathbb{E} R^2 + m^4 = \mathbb{E} R^4 - (\mathbb{E} R^2)^2 + (\mathbb{E} R^2 - m^2)^2 \geq \frac{4}{45} na^4$ by Lemma 4. Thus $4na^2 \text{Var} R \geq \frac{4}{45} na^4$, and we divide by $4na^2 > 0$. $\square$

Corollary 6. $Q_{\sqrt{3}}$ is an isotropic convex body with $\sigma(Q_{\sqrt{3}}) \geq 1/\sqrt{15}$ for every $n \geq 1$. $Q_{1/2}$ has volume 1, barycenter 0 and covariance $\frac{1}{12}I$, and $\sigma(Q_{1/2}) \geq 1/\sqrt{180}$ for every $n \geq 1$.

Proof. By Lemma 3 with $a^2/3 = 1$ (resp. $a = 1/2$, $L^2 = 1/12$), and Theorem 5: $3/45 = 1/15$ and $(1/4)/45 = 1/180$. $\square$

Theorem 7. (a) There are no $C \in \mathbb{R}$ and $N \in \mathbb{N}$ such that $\sigma(K) \leq C n^{-1/4}$ for all $n \geq N$ and all isotropic convex bodies $K \subseteq \mathbb{R}^n$.

(b) There are no C, N with $\sigma_n \leq C n^{-1/4}$ for all $n \geq N$. Hence $\sigma_n = \Theta(n^{-1/4})$ is false, and so is the upper half of the interpolation clause.

(c) Statement (a) also holds in the volume-one convention.

Proof. (a) $C n^{-1/4} \rightarrow 0$, so for some $n \geq \max(N, 1)$ we have $C n^{-1/4} < 1/\sqrt{15} \leq \sigma(Q_{\sqrt{3}})$ by Corollary 6. (b) If $\sigma_n \leq C n^{-1/4}$, then every isotropic K has $\sigma(K) \leq \sigma_n \leq \max(C, 0) n^{-1/4}$, contradicting (a). (c) As (a), with $Q_{1/2}$ and $1/\sqrt{180}$. $\square$

4 Lean formalization

File `lean/Conjecture7783/Basic.lean` (Lean 4, Mathlib v4.33.1; its only import is Mathlib; 340 lines). Definitions: `C7783.IsConvexBody`, `C7783.IsIsotropicConvexBody`, `C7783.IsIsotropicConvexBodyVolOne`, `C7783.thinShell` ($\sqrt{\text{variance } \|\cdot\| \text{ volume}[K]}$), `C7783.sigma` (supremum in $[0, \infty]$), `C7783.cube`. Lemma 2: `isConvexBody_cube`, `volume_cube`, `norm_sq_le_of_mem_cube`. Lemma 3: `setIntegral_cube`, `integral_monomial`, `mean_coord`, `cov_coord`, `fourth_coord`. Lemma 4: `moment_two`, `moment_four`. Theorem 5: `variance_norm_cube`. Corollary 6: `isIsotropic_cube`, `isIsotropicVolOne_cube`. Theorem 7: (a) `C7783.not_thinShell_le_rpow`, (b) `C7783.not_sigma_le_rpow`, (c) `C7783.not_thinShell_le_rpow_volOne`. The statement of (a) in Lean is

$$\neg \exists C : \mathbb{R}, \exists N : \mathbb{N}, \forall n \geq N, \forall K \subseteq \mathbb{R}^n, \text{IsIsotropicConvexBody } K \rightarrow \text{thinShell } K \leq C \cdot n^{-1/4}.$$

Axiom audit. `#print` axioms for each of the three theorems of Theorem 7 reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.